Ilsa B. Ahmed

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EDUCATION

University of Illinois Urbana-Champaign — B.S. Computer Science + Linguistics (CS+X) May 2026

EXPERIENCE

Software Engineer Intern, Lawly — legal-services startup Jun 2026 – Present

- Shipped fixes for 25+ production bugs in a Next.js / React 19 and Fastify + Node.js app on PostgreSQL, including a Stripe checkout error that blocked ~25% of payments.
- Extended the Twilio integration to send SMS appointment reminders to ~2,600 users a week.
- Refactored 20 files of AI-generated TypeScript into typed, tested modules, cutting ~6,000 lines of code and reducing related bug reports by ~30%.

Software Engineer, Telemetry Team, Illini Solar Car Aug 2025 – Dec 2025

- Built a Python asyncio pipeline ingesting 50,000+ CAN bus sensor readings per minute into QuestDB; added indexing and partitioning that cut dashboard query time from ~65 s to ~0.2 s.
- Designed the FastAPI REST layer and React dashboard that 37 team members used to monitor the car live during races and test drives.
- Set up Docker and GitHub Actions CI/CD so every pull request automatically runs tests, linting, and image builds.

Software Engineer Intern, Treevah — pre-seed productivity startup May 2025 – Aug 2025

- Co-built the core 3-panel file manager in React and Node.js, shipped to 172 beta users and featured in the company's investor demo.
- Integrated the Microsoft Graph API to sync nested OneDrive folders, and added Microsoft sign-in with MSAL (OAuth 2.0) on Azure.

Undergraduate Researcher, Forward Data Lab, UIUC Jan 2023 – Apr 2023

- Built document-ranking pipelines with TF-IDF and vector similarity, improving precision and F1 by 50% over the baseline; served results through Flask REST APIs backed by PostgreSQL and MongoDB.

PROJECTS

Stock Bot Battle | C#, ASP.NET Core, PostgreSQL, React, Docker github.com/iahmed20/StockBotBattle

- Built a multiplayer trading game where players write Python bots in a browser editor, trade five stocks against each other in timed rounds, and compete on a live leaderboard.
- Built a matching engine with transactional order execution over an append-only ledger, and ran untrusted user code in resource-limited Docker sandboxes.
- Implemented passwordless magic-link sign-in that stores only hashed, single-use tokens, covered by unit and integration tests against PostgreSQL.

FigureDraw | JavaScript, HTML5 Canvas, CSS iahmed20.github.io/FigureDraw

- Built an in-browser figure-drawing studio with timed practice sessions, layered pencil, ink, and charcoal brushes, and PNG and layered-ZIP export.

SubleaseU | React, Flask, MongoDB, Firebase Auth Team of 3

- Built a sublease marketplace for UIUC students with filtered search and renter-to-poster messaging.
- Built the Flask API, MongoDB schema, and messaging feature, and contributed to the React frontend.

LEADERSHIP & DESIGN

Design Team Member, HackIllinois 2026 — UIUC's student-run hackathon Aug 2025 – Feb 2026

- Designed the illustrations and UI for the 2026 event website (2026.hackillinois.org) in Figma and Illustrator.
- Created the sponsor track intros (Stripe, John Deere, Modal, Caterpillar), the event-day leaderboard display, and campus signage, flyers, and social posts for the HackAstra campaign.

SKILLS

Languages: Python, TypeScript, JavaScript, C#, SQL, Java
Web: React, Next.js, Node.js, ASP.NET Core, FastAPI, Flask, HTML/CSS
Data: PostgreSQL, MongoDB, QuestDB
Cloud & APIs: AWS, Azure, Stripe, Twilio, Microsoft Graph
Tools: Git, Docker, GitHub Actions, Linux, Figma